

ThreatAttest

ThreatAttest

Decentralized threat intelligence on a Cosmos SDK blockchain with a browser extension for real-time protection.

Repository Structure

```
threatattest/
├── chain/                                # Cosmos SDK blockchain node
(threatattestd)
│   ├── cmd/threatattestd/              # Binary entry point
│   ├── x/attestation/                  # Attestation module (publish,
endorse, dispute, revoke)
│   ├── x/identity/                    # Identity module (DNS-bound
tiers, compliance)
│   ├── ante/                          # Ante handler (compliance tier
enforcement)
│   ├── app/                          # App wiring
│   ├── scripts/                      # Build, localnet, protobuf
scripts
│   ├── tests/                        # Shell + JS integration/unit
tests
│   ├── docker/                      # Docker entrypoint + Prometheus
config
│   ├── Makefile                      # Primary chain build targets
│   ├── Dockerfile                   # Multi-stage Docker build
│   └── docker-compose.yml            # Node + monitoring stack
│
├── extension/                          # ThreatAttest Shield browser
extension (MV3)
│   ├── src/                          # Background, content, API, DNS
identity modules
```

```
|   ├── scripts/           # Build, pack, validate scripts
|   ├── manifest.json      # MV3 manifest
|   ├── Makefile           # Extension build targets
|   └── package.json       # NPM scripts + ESLint config
|
└── Makefile               # Root orchestrator (delegates
    to chain/ and extension/)
```

Quick Start

Prerequisites

Tool	Version
Go	≥ 1.21
Node.js	≥ 20
Docker	≥ 24
make	any

Build the chain binary

```
make build
# Output: chain/bin/threatattestd
```

Run tests

```
# All tests (Go unit, shell integration, JS unit)
make test-all
```

```
# Shell integration tests only (no node required)
make test-shell
```

```
# JS unit tests only
make test-js
```

```
# Go unit tests with race detector
make test
```

Start a local devnet

```
make localnet-init      # Initialize chain home, keys, genesis
make localnet-start     # Start node in background
make localnet-status    # Check node status
make localnet-logs      # Tail node logs
make localnet-stop      # Stop node
make localnet-reset     # Wipe and re-initialize
```

Build the browser extension

```
make ext-build          # Production build → extension/dist/
make ext-validate       # Validate manifest + source files
make ext-pack           # Package into extension/build/*.zip
make ext-build-dev      # Development build (localhost API)
```

Docker

```
# Build Docker image
make docker-build

# Run single node container
make docker-run

# Start with Prometheus + Grafana monitoring
make docker-compose-up COMPOSE_PROFILES=monitoring

# Stop
make docker-compose-down
```

Modules

x/attestation

Manages threat attestations on-chain.

Message	Description
MsgPublishAttestation	Publish a new threat record (FILE, URL, IPV4, DOMAIN)
MsgEndorseAttestation	Endorse an existing attestation to boost trust score
MsgDisputeAttestation	File a dispute against an attestation

Message	Description
MsgRevokeAttestation	Revoke your own published attestation
MsgClaimReward	Claim a share of the attestation incentive pool
MsgSubscribe	Lock TATST to activate a paid API tier (PRO/ENTERPRISE)
MsgUnsubscribe	Return locked tokens and revert to the FREE tier

Key fields: `artifact_type`, `artifact_sha256`, `severity`, `confidence`, `ttl_seconds`, `attester_domain`, `attester_selector`

x/identity

DNS-bound identity and compliance tier system.

Tier	Name	Reputation Score	Confidence Multiplier
0	Anonymous	< 100 RS	0.25×
1	Verified	≥ 100 RS	0.50×
2	DNS-Bound	≥ 1000 RS + DNS record	1.00×
3	Expert	≥ 10000 RS + DNSSEC	2.00×

DNS identity: Attesters bind their domain via `_tat.<selector>.<domain>` TXT records. The `attester_domain` + `attester_selector` fields on attestations link to verified DNS identity.

Testing

Shell tests (no node required)

```
cd chain
BINARY=./bin/threatattestd bash tests/test_identity_module.sh
BINARY=./bin/threatattestd bash tests/test_attester_domain.sh
BINARY=./bin/threatattestd bash tests/test_compliance_tiers.sh
```

When a node is not running, integration sections are gracefully skipped. All CLI-structural and unit tests run offline.

JS unit tests (151 tests)

```
node chain/tests/test_extension_badges.js
```

Covers all pure functions in `extension/src/dns_identity.js`:
`buildIdentityBadge`, `parseIdentityFlags`, `normalizeDomain`,
`resolveIdentity`, `hasActiveIdentity`, cache behaviour, DOM
contracts, and stress cases.

Extension validation

```
cd extension && node scripts/validate.js
```

Checks manifest validity, MV3 fields, all referenced files exist, JS
syntax, version consistency, and `dns_identity.js` export contract.

CI/CD

GitHub Actions workflows in `chain/.github/workflows/`:

Workflow	Triggers	Jobs
<code>ci.yml</code>	push, PR	lint, test, build, test-shell, test-js, validate-ext
<code>release.yml</code>	<code>v*.*.*</code> tags	goreleaser (multi- arch), ext-release

Release artifacts

GoReleaser produces: - `threatattestd_linux_amd64.tar.gz` (CGO
enabled) - `threatattestd_linux_arm64.tar.gz` -
`threatattestd_darwin_amd64.tar.gz` -
`threatattestd_darwin_arm64.tar.gz` -
`threatattestd_windows_amd64.zip` - `checksums.txt` - Docker
manifests → ghcr.io/threatattest/threatattestd

Extension release: - `threatattest-shield-<version>.zip` attached to
GitHub Release

Development

Adding a new attestation field

1. Add field to `x/attestation/types/types.go`
2. Update `MsgPublishAttestation` in `x/attestation/messages/messages.go`

3. Add CLI flag to `cmd/threatattestd/cmd/root.go`
4. Add validation in `x/attestation/keeper/msg_server.go`
5. Update `extension/src/api.js` to surface the field
6. Add tests to `chain/tests/test_attester_domain.sh`

Protobuf regeneration

```
make proto-gen    # Requires buf CLI or Docker
make proto-lint   # Lint proto files
```

Linting

```
make lint         # golangci-lint (Go) + ESLint (JS)
make lint-fix     # Auto-fix where possible
```

License

Apache 2.0 — see [LICENSE](#)